# Filing Rate Decomposition and Back-Rent Analysis

**Philadelphia Evictions, 2011–2019**

2026-03-27

## Table of contents


**1 Overview** **1**

**2 Part 0: Entity Consolidation** **2**
2.1 Two-phase owner linkage . . . 2
2.2 PID × year coverage . . . 2
2.3 Largest entities and conglomerates by category . . . 2

**3 Part A: Cross-Sectional Filing Rate Decomposition** **4**
3.1 A1: Descriptive means by owner type and portfolio . . . 4
3.2 A1b: Owner-Type Bridge . . . 6
3.3 A2: Sequential $R^2$ Decomposition . . . 6
3.4 A3: OLS Coefficient Table (M1–M5) . . . 7
3.5 A4: Market Concentration in ZIP x Filing-Rate Markets . . . 9
3.6 A5: Market-Year Rent Regressions . . . 10
3.7 A6: Alternative Market Definitions . . . 10

**4 Part B: Case-Level Back-Rent Analysis** **12**
4.1 B0: Sample . . . 12
4.2 B1: Descriptive statistics . . . 13
4.3 B2: Months-of-Arrears Regressions . . . 13
4.4 B3: Back-Rent Elasticity . . . 14
4.5 B4: Racial Disparity Regressions . . . 14

**5 Summary** **18**
5.1 Caveats . . . 18


## 1 Overview

This report summarizes two analyses of the Philadelphia eviction record (2011–2019):

1. **Entity consolidation** (`build-ownership-panel.R`): A two-phase owner linkage pipeline mapping RTT transfers and OPA parcels to legal entities and conglomerates.
2. **Filing rate decomposition** (`analyze-filing-decomposition.R`):
   - *Part A*: What fraction of cross-sectional variation in building-level filing rates is attributable to building characteristics, neighborhood, tenant composition, and landlord type?
   - *Part B*: Do corporate or high-filing landlords file evictions with less accumulated back rent? Is the plaintiff (often a management company) different from the owner?

**Commercial case filter (Part B):** Commercial cases are identified by the defendant in position 1 of the party-names table, not by the self-reported `commercial` flag in the evictions data (which is unreliable). Counterclaim parties (e.g. a corporate landlord counter-sued by a tenant) are excluded from the defendant classification. This is produced by `impute-race-preflight.R` (`eviction_case_commercial` product).

**Sample restrictions (Part B):** non-commercial cases, `ongoing_rent` 300–6000, `total_rent` 300–15000, `months >= 1`, single-parcel PID match, 2011–2019.

---

# 2 Part 0: Entity Consolidation

## 2.1 Two-phase owner linkage

Phase 1 normalizes raw RTT grantee names (stripping entity suffixes, roman numerals, bank/trust fragments) and groups exact-match normalized names into **entities**. Persons and trusts are grouped by *(name, PID)* rather than name alone, preventing common names from forming spurious hubs.

Phase 2 uses a name+address self-join on OPA mailing data: corporations match on *(business_short_name, address)* where `business_short_name` further strips functional descriptor words (PROPERTIES, MANAGEMENT, REALTY, etc.); persons/trusts match on *(exact name, address)*. A registered-agent filter drops addresses linked to more than 300 distinct entities. Connected components of the resulting name+address graph form **conglomerates**.

Phase 3 adds one-shot RTT→OPA transitive edges (last RTT grantee vs. 2024 OPA owner for each PID), subject to a name-similarity gate: corporations must share the same `business_short_name` ( 4 chars); persons must share 1 token of 5 characters. This prevents government entities (City of Philadelphia, PHA) from absorbing hundreds of unrelated parcels acquired by tax-exempt conveyances not captured in RTT.

## 2.2 PID × year coverage

## 2.3 Largest entities and conglomerates by category

---

Table 1: Entity consolidation summary

| Stage | Count |
|---|---|
| Phase 1 entities (RTT exact-name match) | 451,346 |
| OPA-only entity additions | 262,935 |
| Total entities | 714,281 |
| Registered-agent addresses dropped (>300 entities each) | 44 (all SIMPLIFILE LC E-RECORDING variants) |
| Entities entering mailing-address graph | 64,461 |
| Conglomerates formed | 49,651 |
| Multi-entity conglomerates | 3,063 |
| Phase 1 entities merged by Phase 2 | 4,464 |
| Name lookup entries (all sources) | 719,215 |

Table 2: PID x year ownership panel coverage

| Metric | Value |
|---|---|
| PID x year panel rows | 14,015,256 (583,969 PIDs x 24 years) |
| RTT-sourced owner rows | 5,128,881 (36.6%) |
| OPA-fallback owner rows | 8,886,375 (63.4%) |
| Entity ID coverage | 14,015,256 (100%) |
| Portfolio bin: 1 PID | 3,522,595 (43.9% of 2011-19 PID-yrs) |
| Portfolio bin: 2–4 PIDs | 981,376 (12.2%) |
| Portfolio bin: 5–19 PIDs | 425,827 (5.3%) |
| Portfolio bin: 20+ PIDs | 325,923 (4.1%) |

Table 3: Largest sample entities by owner category (top two per category).

| Category | Rank | Primary name | Entity ID | Max PIDs | Max units | Sample PIDs | Entities in CG |
|---|---|---|---|---|---|---|---|
| Financial-Intermediary | 1 | FEDERAL NATIONAL MORTGAGE ASSOCIATION | 117 | 343 | 416 | 1,285 | 1 |
| Financial-Intermediary | 2 | WELLS FARGO BANK NA | 1544 | 225 | 260 | 428 | 1 |
| For-profit corp | 1 | WEST VILLAGE GROUP LLC | 15168 | 87 | 755 | 87 | 2 |
| For-profit corp | 2 | VDBE WOODCREST LLC | 50564 | 20 | 696 | 20 | 1 |
| Government-Federal | 1 | UNITED STATES OF AMERICA INC | 576683 | 26 | 53 | 26 | 1 |
| Government-Federal | 2 | ADMIN OF VETERAN AFFAIRS | 580202 | 11 | 11 | 11 | 1 |
| Government-Local | 1 | CITY OF PHILA | 566959 | 1,655 | 1,215 | 1,655 | 293 |
| Government-Local | 2 | SEPTA | 810245 | 278 | 820 | 278 | 1 |
| Nonprofit | 1 | TEMPLE UNIV COMMONWEALTH SYSTEM OF HIGHER ED | 576224 | 112 | 995 | 112 | 1 |
| Nonprofit | 2 | GEDR UPENN LLC | 576729 | 146 | 757 | 146 | 42 |
| PHA | 1 | HLP PHILADELPHIA HOUSING | 4796 | 5,543 | 9,883 | 5,544 | 30 |
| PHA | 2 | PHILADELPHIA HOUSING AUTHORITY DEVELOPMENT CORPORATION | 22597 | 4 | 61 | 4 | 30 |
| Person | 1 | DOMB ALLAN | 140314 | 226 | 548 | 226 | 34 |
| Person | 2 | SUMMIT PARK EAST | 825045 | 2 | 487 | 2 | 1 |
| Religious | 1 | EVANGELICAL MANOR | 654110 | 1 | 159 | 1 | 1 |
| Religious | 2 | TIOGA PRESBYTERIAN APARTMENTS INC | 576320 | 3 | 116 | 3 | 1 |
| Trust | 1 | BIENENFELD JACK TR JACK BIENENFELD SEPARATE SHARE TRUST UND | 565889 | 2 | 108 | 2 | 1 |
| Trust | 2 | 2011 IRREVOCABLE TRUST FOR FAMILY OF HEN BIENENFELD JACK TR | 562833 | 1 | 74 | 1 | 1 |

Table 4: Largest sample conglomerates by primary owner category (top two per category).

| Category | Rank | Primary name | CG ID | Primary entity ID | Entities in CG | Max PIDs | Max units |
|---|---|---|---|---|---|---|---|
| Financial-Intermediary | 1 | FEDERAL NATIONAL MORTGAGE ASSOCIATION | 117 | 117 | 1 | 343 | 416 |
| Financial-Intermediary | 2 | WELLS FARGO BANK NA | 1544 | 1544 | 1 | 225 | 260 |
| For-profit corp | 1 | PARK BRADFORD APARTMENTS | 1028 | 573283 | 17 | 72 | 1,011 |
| For-profit corp | 2 | FERNROCK APARTMENTS 2 LP | 3203 | 568708 | 53 | 206 | 983 |
| Government-Federal | 1 | UNITED STATES OF AMERICA INC | 576683 | 576683 | 1 | 26 | 53 |
| Government-Federal | 2 | ADMIN OF VETERAN AFFAIRS | 580202 | 580202 | 1 | 11 | 11 |
| Government-Local | 1 | CITY OF PHILA | 1 | 566959 | 293 | 5,496 | 2,700 |
| Government-Local | 2 | SEPTA | 810245 | 810245 | 1 | 278 | 820 |
| Nonprofit | 1 | GEDR UPENN LLC | 1303 | 576729 | 42 | 255 | 1,586 |
| Nonprofit | 2 | TEMPLE UNIV COMMONWEALTH SYSTEM OF HIGHER ED | 576224 | 576224 | 1 | 112 | 995 |
| PHA | 1 | HLP PHILADELPHIA HOUSING | 4796 | 4796 | 30 | 7,462 | 10,384 |
| PHA | 2 | PHILADELPHIA HOUSING ASSOCIATES LLC | 6718 | 6718 | 1 | 3 | 6 |
| Person | 1 | DOMB ALLAN | 986 | 140314 | 34 | 302 | 929 |
| Person | 2 | SUMMIT PARK EAST | 825045 | 825045 | 1 | 2 | 487 |
| Religious | 1 | TEMPLE VILLAS II LLC | 13128 | 15134 | 39 | 67 | 242 |
| Religious | 2 | EVANGELICAL MANOR | 654110 | 654110 | 1 | 1 | 159 |
| Trust | 1 | BIENENFELD JACK TR JACK BIENENFELD SEPARATE SHARE TRUST UND | 565889 | 565889 | 1 | 2 | 108 |
| Trust | 2 | 2011 IRREVOCABLE TRUST FOR FAMILY OF HEN BIENENFELD JACK TR | 562833 | 562833 | 1 | 1 | 74 |

# 3 Part A: Cross-Sectional Filing Rate Decomposition

**Sample:** ever-rental buildings in 2011–2019 with non-missing EB filing rate, zip, census tract, and `rental_share`. N reported in table below. No minimum unit count restriction — if a building appears in eviction filings it is treated as a rental.

**Rental share correction:** The EB filing rate denominator includes all observed unit-years, but single-family homes are sometimes rented only intermittently, deflating their apparent filing rates. All models (M1 onward) include `rental_share` = fraction of sample years with an active rental license, correcting for this. Philadelphia Housing Authority (PHA) buildings are exempt from the City licensing requirement; their `rental_share` is imputed to 1 (always fully rented).

## 3.1 A1: Descriptive means by owner type and portfolio

Nonprofits stand out with a mean EB rate of 0.18 — far above the other types and likely reflecting a small set of high-density affordable-housing buildings with concentrated poverty. PHA-owned buildings also exceed person-owned buildings (0.10 vs. 0.05). For-profit corporations file at 0.07, modestly above the 0.05 person baseline.

Table 5: EB filing rate by owner type and portfolio (cross-section, 2011–2019). Mean units owned = avg total units across all properties in the landlord's portfolio. Portfolio rows exclude PHA and government owners.

| Group | N (PIDs) | Mean units owned | Mean EB rate | Median EB rate |
|---|---|---|---|---|
| **Owner type (Person = reference)** | | | | |
| Owner: Person | 140,388 | 4 | 0.040 | 0.005 |
| Owner: For-profit corp | 40,023 | 26 | 0.064 | 0.005 |
| Owner: Nonprofit | 1,110 | 110 | 0.170 | 0.131 |
| Owner: PHA | 5,471 | 9,593 | 0.011 | 0.005 |
| Owner: Religious | 425 | 15 | 0.035 | 0.005 |
| Owner: Trust | 810 | 6 | 0.061 | 0.005 |
| Owner: Government-Local | 372 | 384 | 0.034 | 0.005 |
| Owner: Government-Federal | 6 | 29 | 0.007 | 0.004 |
| Owner: Financial-Intermediary | 949 | 46 | 0.055 | 0.005 |
| **Entity portfolio (non-PHA / non-government)** | | | | |
| Entity: 1 prop. | 139,913 | 3 | 0.041 | 0.005 |
| Entity: 2-4 prop. | 21,451 | 7 | 0.051 | 0.005 |
| Entity: 5-19 prop. | 13,730 | 16 | 0.068 | 0.005 |
| Entity: 20+ prop. | 8,611 | 112 | 0.086 | 0.008 |
| **Conglomerate portfolio (non-PHA / non-government)** | | | | |
| Cong.: 1 prop. | 123,689 | 4 | 0.039 | 0.005 |
| Cong.: 2-4 prop. | 28,771 | 6 | 0.048 | 0.005 |
| Cong.: 5-19 prop. | 17,762 | 17 | 0.064 | 0.005 |
| Cong.: 20+ prop. | 13,483 | 183 | 0.086 | 0.007 |

Table 6: Race-augmented version of Table 5. Building Black share is unit-weighted across buildings; case Black probability is case-weighted across linked non-commercial cases with race imputation. Portfolio rows exclude PHA and government owners.

| Group | N (PIDs) | Mean units owned | Mean EB rate | Mean bldg. Black share | Mean case Black prob. |
|---|---|---|---|---|---|
| **Owner type (Person = reference)** | | | | | |
| Owner: Person | 140,388 | 4 | 0.040 | 32.7% | 54.2% |
| Owner: For-profit corp | 40,023 | 26 | 0.064 | 33.5% | 56.0% |
| Owner: Nonprofit | 1,110 | 110 | 0.170 | 53.9% | 76.8% |
| Owner: PHA | 5,471 | 9,593 | 0.011 | 73.9% | 79.1% |
| Owner: Religious | 425 | 15 | 0.035 | 46.9% | 81.0% |
| Owner: Trust | 810 | 6 | 0.061 | 30.4% | 44.7% |
| Owner: Government-Local | 372 | 384 | 0.034 | 17.0% | 45.3% |
| Owner: Government-Federal | 6 | 29 | 0.007 | 41.4% | 2.7% |
| Owner: Financial-Intermediary | 949 | 46 | 0.055 | 47.7% | 63.5% |
| **Entity portfolio (non-PHA / non-government)** | | | | | |
| Entity: 1 prop. | 139,913 | 3 | 0.041 | 31.2% | 52.1% |
| Entity: 2-4 prop. | 21,451 | 7 | 0.051 | 34.5% | 59.9% |
| Entity: 5-19 prop. | 13,730 | 16 | 0.068 | 43.6% | 61.0% |
| Entity: 20+ prop. | 8,611 | 112 | 0.086 | 52.7% | 74.2% |
| **Conglomerate portfolio (non-PHA / non-government)** | | | | | |
| Cong.: 1 prop. | 123,689 | 4 | 0.039 | 31.2% | 52.4% |
| Cong.: 2-4 prop. | 28,771 | 6 | 0.048 | 33.1% | 55.2% |
| Cong.: 5-19 prop. | 17,762 | 17 | 0.064 | 39.1% | 57.5% |
| Cong.: 20+ prop. | 13,483 | 183 | 0.086 | 43.9% | 66.9% |

### 3.2 A1b: Owner-Type Bridge

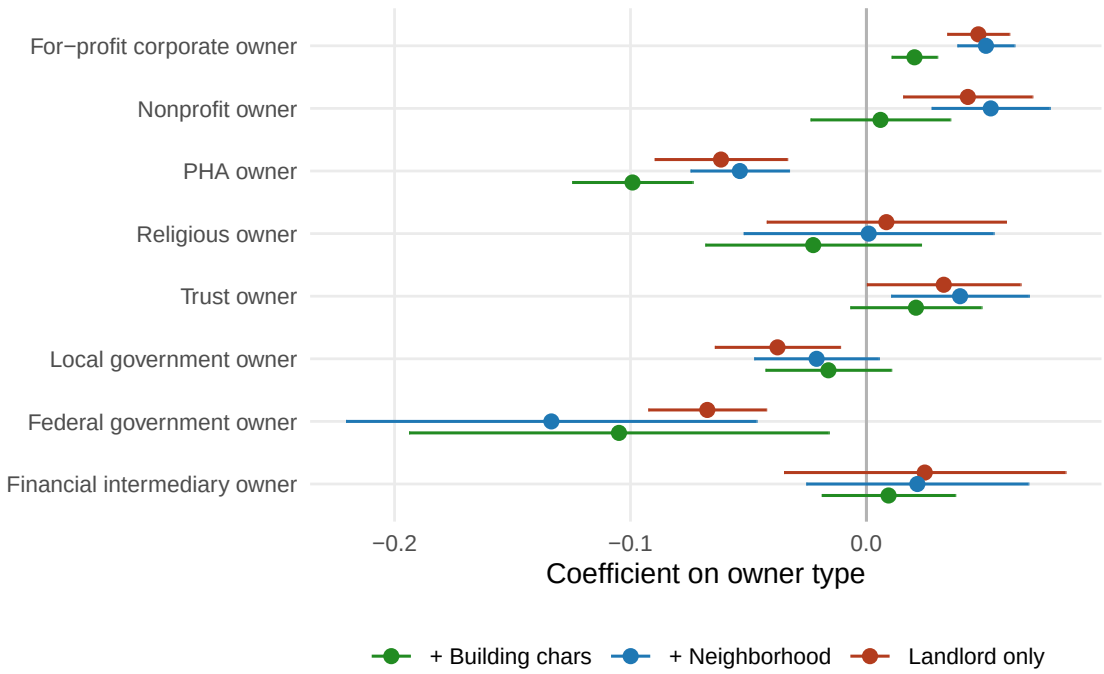


Figure 1: Owner-type coefficients as controls are added. Reference group: person owners. Error bars show 95% confidence intervals.

The nonprofit coefficient falls sharply as neighborhood and then building controls enter. That is the core reason the unconditional mean in Table 5 is so much larger than the conditional owner-type coefficient in Table 7: nonprofits disproportionately own larger conglomerate portfolios and large multifamily buildings in higher-filing tracts, but once those comparisons are made within tract/building cells, the residual nonprofit-owner effect is small.

### 3.3 A2: Sequential $R^2$ Decomposition

Six nested OLS models, weighted by `total_units`, outcome = `filing_rate_eb_pre_covid`. $\Delta R^2$ isolates the variance attributable to each group of factors added. M4 and M5 both branch from M3: M4 uses entity-level portfolio (per LLC/name); M5 uses conglomerate-level portfolio (across all linked LLCs under common ownership, from improved Phase 2/3 reconciliation). M5's $\Delta R^2$ is relative to M3.

**Interpretation.** Geography dominates the decomposition. The $\Delta R^2$ for the ZIP→tract upgrade alone (M2→M3) is roughly 9× the landlord component ($\Delta R^2 \approx 0.061$ vs 0.007). The total geographic contribution — from no geographic controls (M0) through building characteristics (M1) to ZIP FE (M2) to tract FE (M3) — is substantially larger still: $\Delta R^2 \approx 0.208$ for all geographic FE vs $\approx 0.007$ for landlord type, a roughly 30× ratio. Tenant racial composition adds a further increment beyond tract FE, suggesting neighborhood sorting by race is a distinct channel from simple geographic sorting. The `rental_share` covariate (included from M1) corrects for the deflated

Table 7: Sequential $R^2$ decomposition. Between-neighborhood raw variance share: 0.118.

| Model | Description | $R^2$ | Adj. $R^2$ | $\Delta R^2$ | N |
|---|---|---|---|---|---|
| M0 | Intercept only | — | — | — | 188,435 |
| M1 | Zip FE + building chars + rental share | 0.1467 | 0.1463 | — | 188,426 |
| M2 | Tract FE + building chars + rental share | 0.2081 | 0.2063 | 0.0614 | 188,432 |
| M3 | Tract FE + building + pct_black + rental share | 0.2190 | 0.2171 | 0.0109 | 174,904 |
| M4 | Tract FE + building + pct_black + rental share + landlord (entity portfolio) | 0.2300 | 0.2281 | 0.0110 | 174,904 |
| M5 | Tract FE + building + pct_black + rental share + landlord (conglomerate portfolio) | 0.2312 | 0.2292 | 0.0121 | 174,904 |

*Note:* Weighted by mean total units. Cluster SE at tract.

filing rates of intermittently-rented properties; its $\Delta R^2$ contribution is absorbed into M1 and is not separately isolated. The landlord-type effect is small but present; numbers updated after re-run.

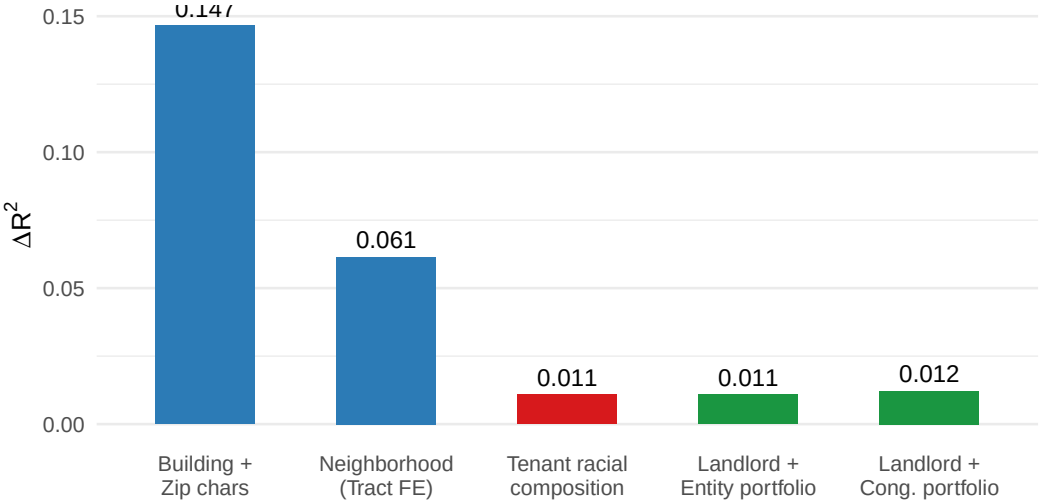


Figure 2: Incremental R-squared by component group. Outcome: EB filing rate. Weighted by unit count. Portfolio bars show delta from M3 for entity (M4) and conglomerate (M5) portfolio measures.

## 3.4 A3: OLS Coefficient Table (M1–M5)

This table reorganizes the owner-type comparison to match the bridge plot: first landlord characteristics only, then adding neighborhood controls, then building controls on the same common conglomerate-portfolio sample. Neighborhood controls now include both mean Black share and mean female share. Entries after the first two columns are owner-type coefficients relative to person owners, with clustered standard errors in parentheses.

Table 8: Owner-Type Bridge Regressions

| Dependent Variable: | filing_rate_eb_pre_covid | | |
|---|---|---|---|
| Model: | (1) | (2) | (3) |
| *Variables* | | | |
| Constant | 0.0417*** | | |
| | (0.0018) | | |
| For-profit corporate owner | 0.0474*** | 0.0507*** | 0.0204*** |
| | (0.0067) | (0.0062) | (0.0049) |
| Nonprofit owner | 0.0430*** | 0.0527*** | 0.0060 |
| | (0.0140) | (0.0128) | (0.0151) |
| PHA owner | -0.0616*** | -0.0537*** | -0.0991*** |
| | (0.0143) | (0.0106) | (0.0130) |
| Religious owner | 0.0085 | 0.0010 | -0.0226 |
| | (0.0259) | (0.0270) | (0.0233) |
| Trust owner | 0.0328** | 0.0397*** | 0.0210 |
| | (0.0166) | (0.0149) | (0.0142) |
| Local government owner | -0.0377*** | -0.0211 | -0.0161 |
| | (0.0135) | (0.0135) | (0.0136) |
| Federal government owner | -0.0674*** | -0.1335*** | -0.1049** |
| | (0.0127) | (0.0444) | (0.0454) |
| Financial intermediary owner | 0.0247 | 0.0216 | 0.0094 |
| | (0.0304) | (0.0240) | (0.0144) |
| Conglomerate portfolio: 2-4 properties | 0.0056 | 0.0033 | -0.0010 |
| | (0.0067) | (0.0063) | (0.0055) |
| Conglomerate portfolio: 5-19 properties | 0.0083 | 0.0023 | -0.0022 |
| | (0.0067) | (0.0051) | (0.0042) |
| Conglomerate portfolio: 20+ properties | 0.0316** | 0.0271** | 0.0219** |
| | (0.0145) | (0.0110) | (0.0104) |
| Mean Black share | | 0.0463*** | 0.0351*** |
| | | (0.0053) | (0.0050) |
| Mean Female share | | 0.0097*** | 0.0017 |
| | | (0.0025) | (0.0023) |
| 2-5 units | | | 0.0011 |
| | | | (0.0019) |
| 21-50 units | | | 0.0237*** |
| | | | (0.0091) |
| 51+ units | | | 0.0202* |
| | | | (0.0103) |
| 6-20 units | | | -0.0035 |
| | | | (0.0062) |
| Rental-share correction | | | 0.0527*** |
| | | | (0.0035) |
| *Fixed-effects* | | | |
| Tract | | Yes | Yes |
| Building type | | | Yes |
| Year built decade | | | Yes |
| Quality grade | | | Yes |
| *Fit statistics* | | | |
| Observations | 172,871 | 172,870 | 172,870 |
| $R^2$ | 0.03558 | 0.15841 | 0.23210 |
| Adjusted $R^2$ | 0.03551 | 0.15648 | 0.23012 |

*Clustered (Tract) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

### 3.5 A4: Market Concentration in ZIP x Filing-Rate Markets

To mirror the consolidation analysis from the other project, I construct a landlord market-year panel directly from the ownership panel. A **market** is defined as:

[ market = ZIP code $\times 1\{\text{building EB filing rate} > 0.05\}$. ]

This creates two submarkets within each ZIP: a higher-filing segment and a lower-filing segment. Within each market-year, landlord concentration is measured using conglomerate-level unit shares, so linked LLCs count as one owner. The main concentration metric is unit-based HHI on the standard 0–10,000 scale.

Table 9: ZIP x filing-rate market concentration summary in 2011 and 2019.

| Segment | Year | Markets | Units | Mean HHI | Top-1 share (%) | Top-3 share (%) | Corp. unit share (%) | Median owners |
|---|---|---|---|---|---|---|---|---|
| Lower-filing ZIP segment | 2011 | 50 | 248,916 | 113 | 6.2 | 10.8 | 35.9 | 1916 |
| Lower-filing ZIP segment | 2019 | 51 | 272,511 | 112 | 5.9 | 10.6 | 40.0 | 1887 |
| Lower-filing ZIP segment | 2011 | 52 | 97,893 | 203 | 7.2 | 14.9 | 43.3 | 518 |
| Lower-filing ZIP segment | 2019 | 53 | 101,381 | 209 | 7.2 | 15.5 | 48.5 | 496 |

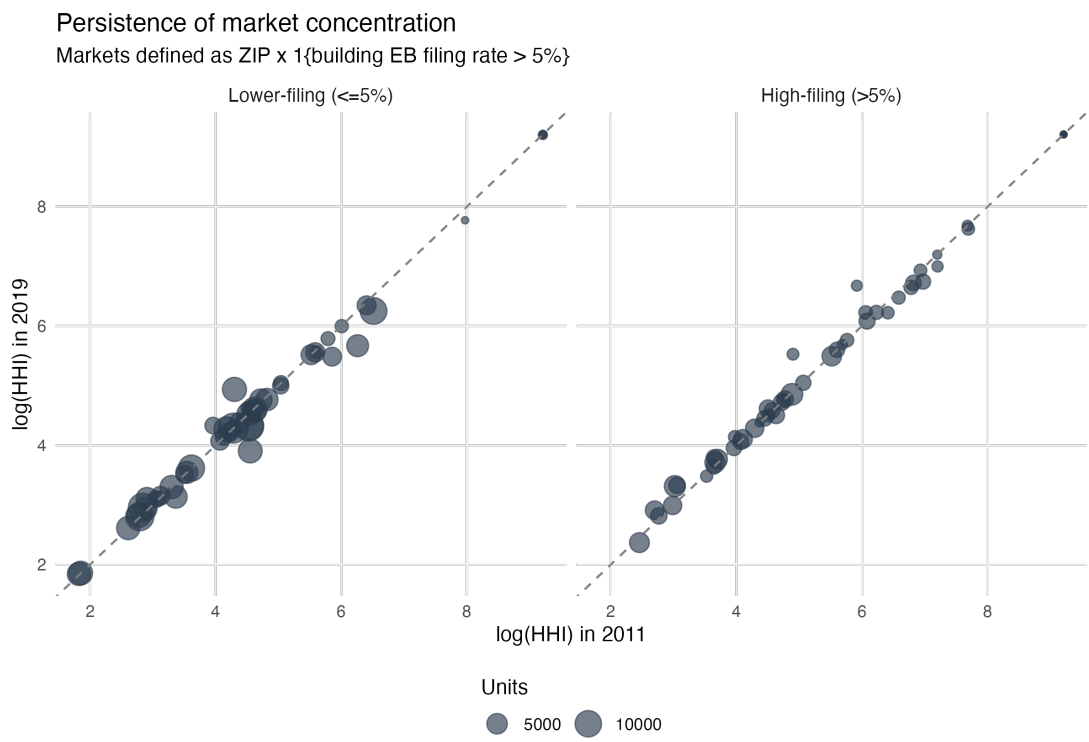


Figure 3: Persistence of log(HHI) between the first and last sample years. Each point is a ZIP x filing-rate market, sized by market units.

Across the sample, the higher-filing ZIP segments are consistently more concentrated and more corporate-owned than the lower-filing segments. The persistence plot shows that this is not just cross-sectional noise: markets that start concentrated generally stay concentrated through the end of the sample.

## 3.6 A5: Market-Year Rent Regressions

These regressions parallel the old consolidation script, but use the Philadelphia building panel. The outcome is the unit-weighted mean of building `log_med_rent` within each ZIP x filing-rate market-year.

Table 10: Market Concentration and Rent in ZIP x Filing-Rate Markets

| Dependent Variable: | mean_log_rent | | | | |
|---|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) | (5) |
| *Variables* | | | | | |
| Log HHI | 0.0250 | -0.0051 | -0.0259 | -0.0280 | 0.0069 |
| | (0.0270) | (0.0376) | (0.0818) | (0.0829) | (0.0674) |
| Corporate unit share (%) | | 0.0031 | 0.0074*** | 0.0074*** | 0.0060** |
| | | (0.0029) | (0.0026) | (0.0026) | (0.0025) |
| Mean log income | | | | -0.0190 | -0.0365 |
| | | | | (0.0279) | (0.0281) |
| Log market units | | | | | 0.2290** |
| | | | | | (0.1103) |
| *Fixed-effects* | | | | | |
| Year | Yes | Yes | Yes | Yes | Yes |
| market_id | | | Yes | Yes | Yes |
| *Fit statistics* | | | | | |
| Observations | 869 | 869 | 865 | 860 | 860 |
| $R^2$ | 0.07468 | 0.08424 | 0.90592 | 0.90597 | 0.90675 |
| Adjusted $R^2$ | 0.06499 | 0.07357 | 0.89248 | 0.89245 | 0.89320 |

*Clustered (market_cluster) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The identifying variation in columns (3)–(5) is within market over time, after absorbing market and year fixed effects. In other words, these columns ask whether a given ZIP x filing-rate segment becomes more concentrated in years when it also becomes more expensive.

## 3.7 A6: Alternative Market Definitions

Two easy robustness checks use alternative market definitions: (i) census tracts, and (ii) ZIP code crossed with building-size bins `1`, `2--4`, `5--49`, and `50+` units.

Table 11: Tract-market concentration summary in the first and last sample years.

| Year | Markets | Units | Mean HHI | Top-1 share (%) | Top-3 share (%) | Corp. unit share (%) | Median owners |
|---|---|---|---|---|---|---|---|
| 2011 | 384 | 346,775 | 375 | 11.2 | 20.6 | 38.0 | 400 |
| 2019 | 384 | 373,854 | 371 | 11.1 | 20.6 | 42.3 | 411 |

Table 12: ZIP x property-size market concentration summary in the first and last sample years.

| Size bin | Year | Markets | Units | Mean HHI | Top-1 share (%) | Top-3 share (%) | Corp. unit share (%) | Median owners |
|---|---|---|---|---|---|---|---|---|
| 1 | 2011 | 52 | 138,190 | 55 | 3.7 | 5.3 | 23.7 | 1868 |
| 1 | 2019 | 52 | 140,435 | 54 | 3.7 | 5.3 | 26.6 | 1847 |
| 2-4 | 2011 | 48 | 89,766 | 43 | 3.2 | 5.1 | 26.4 | 676 |
| 2-4 | 2019 | 50 | 94,030 | 40 | 3.2 | 5.2 | 30.5 | 654 |
| 5-49 | 2011 | 49 | 67,264 | 276 | 7.7 | 15.9 | 57.7 | 63 |
| 5-49 | 2019 | 49 | 78,078 | 231 | 6.9 | 14.5 | 60.2 | 63 |
| 50+ | 2011 | 47 | 51,589 | 1185 | 19.4 | 43.1 | 70.8 | 8 |
| 50+ | 2019 | 48 | 61,349 | 1085 | 18.4 | 39.5 | 73.5 | 8 |

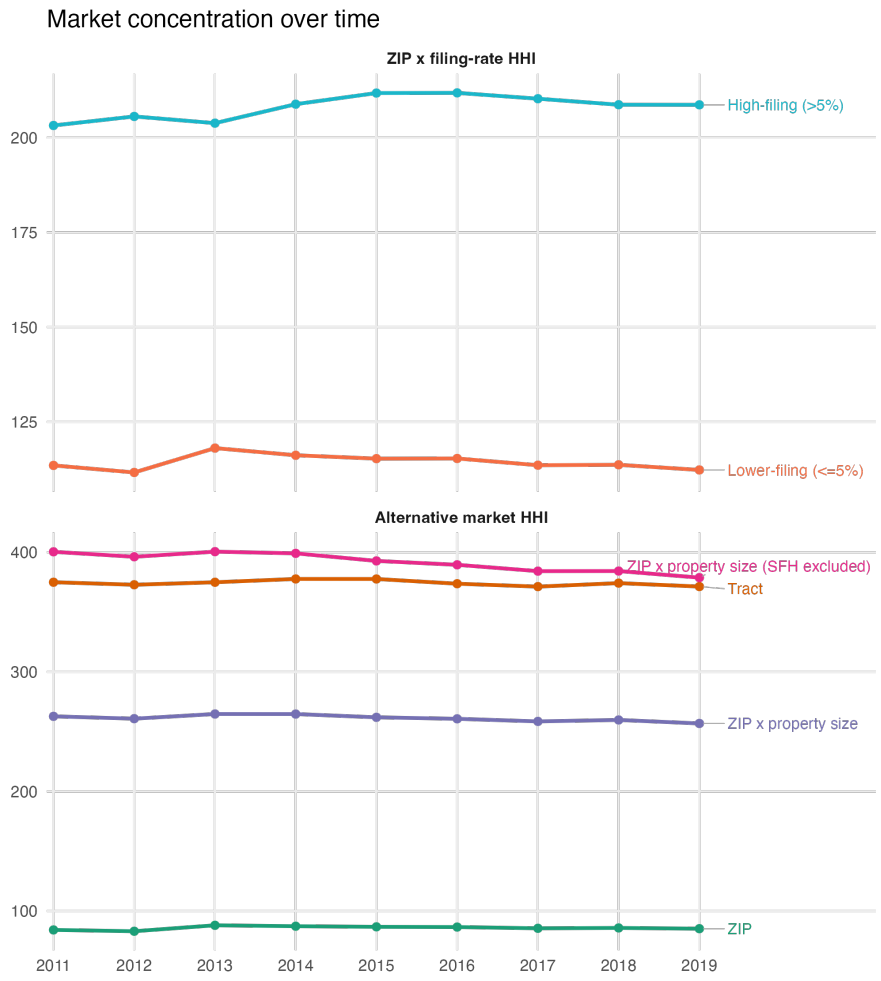


Figure 4: HHI over time in ZIP x filing-rate markets and alternative market definitions. End-of-line labels use the plotted series names.

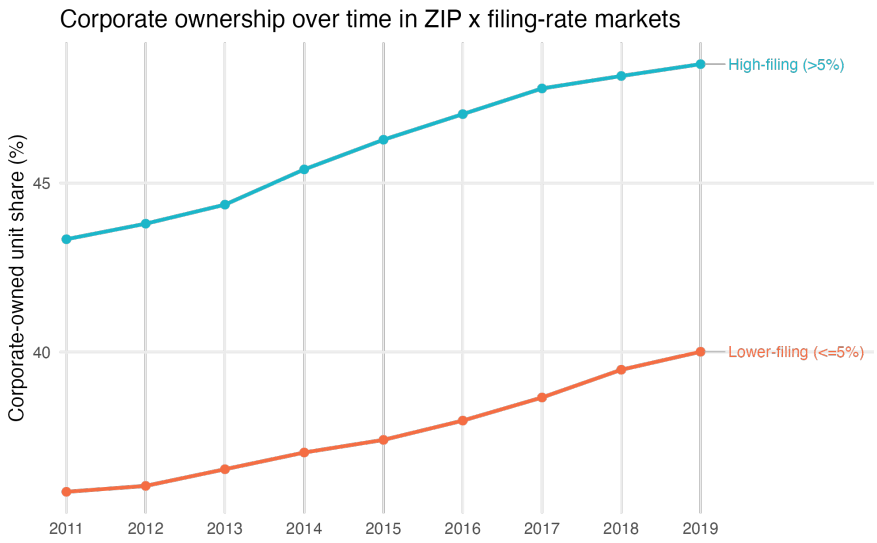


Figure 5: Corporate-owned unit share over time in ZIP x filing-rate markets. End-of-line labels use the plotted series names.

---

# 4 Part B: Case-Level Back-Rent Analysis

## 4.1 B0: Sample

Table 13: Case-level sample construction

| Step | Value |
|---|---|
| Eviction cases 2011–2019 (before commercial filter) | 207,244 |
| Commercial cases dropped (defendant #1 is a business entity) | 5,669 |
| Non-commercial cases retained | 201,575 |
| After rent/months filters (ongoing > 0, months >= 1) | 163,644 |
| After PID linkage (unique-parcel match) | 153,746 (76.3%) |
| After rent winsorization (ongoing 300–6000, total 300–15000) | 152,733 |
| Entity ID (building owner) linked | 95.5% |
| Plaintiff in name lookup | 29.1% |
| Plaintiff is building owner (exact entity match) | 15.8% |
| — Plaintiff type breakdown (winsorized sample) — | |
| For-profit corporate plaintiff | 51.5% (78,615) |
| Philadelphia Housing Authority (PHA) plaintiff | 4.1% (6,317) |
| Nonprofit / affordable-housing plaintiff | 2.3% (3,518) |
| Individual plaintiff | 42.1% (64,283) |
| — Owner type breakdown — | |
| Person owner | 48.1% (73,467) |
| For-profit corp owner | 46.6% (71,186) |
| PHA owner | 3.6% (5,560) |
| Nonprofit owner | 1.7% (2,520) |

Only 15.8% of cases have a plaintiff matching the building owner entity exactly, confirming that management companies file the majority of evictions.

**Note on PHA plaintiff cases:** 6,317 PHA plaintiff cases survive the rent filter (4.1% of the winsorized sample). PHA income-based rents have a median of $221/month and roughly 78% fall below the $300 rent floor; the surviving cases represent above-poverty-line PHA tenants in higher-rent units.

**Note on PHA owner type:** 5,560 cases are in PHA-owned buildings (3.6%). The PHA-owned flag uses entity-ID matching through the name-lookup table, so it identifies the subset of cases where the building is formally owned by the Housing Authority regardless of who filed.

## 4.2 B1: Descriptive statistics

Table 14: Back-rent descriptives by landlord/plaintiff type and landlord scale (winsorized sample). Mean cong. units = mean total units across the conglomerate portfolio in the case year.

| Category | N | Med. months | Mean rent | Mean cong. units | Pct. win |
|---|---|---|---|---|---|
| **By plaintiff/owner type** | | | | | |
| Non-fp-corp plaintiff | 73,978 | 2.10 | $730 | 732 | 5.3% |
| For-profit corp plaintiff | 79,999 | 2.00 | $838 | 232 | 2.0% |
| Non-PHA plaintiff | 147,651 | 2.00 | $803 | 143 | 3.7% |
| PHA plaintiff | 6,326 | 1.72 | $391 | 8025 | 1.8% |
| Non-nonprofit plaint. | 150,437 | 2.00 | $788 | 485 | 3.6% |
| Nonprofit plaintiff | 3,540 | 2.12 | $716 | 275 | 1.3% |
| Person owner | 64,784 | 2.14 | $765 | 15 | 5.6% |
| Corporate owner | 81,981 | 2.00 | $794 | 848 | 2.1% |
| Non-PHA owner | 148,375 | 2.00 | $800 | 103 | 3.6% |
| PHA owner | 5,602 | 1.79 | $408 | 9987 | 1.7% |
| Mgmt co. filer | 125,000 | 2.00 | $792 | 200 | 4.0% |
| Owner-filer | 28,977 | 2.00 | $761 | 1619 | 1.9% |
| High-filing bldg | 80,721 | 2.00 | $795 | 188 | 2.6% |
| Lower-filing bldg | 61,874 | 2.19 | $765 | 843 | 5.0% |
| **By landlord scale (units outside this building)** | | | | | |
| Scale: single-property | 71,764 | 2.01 | $816 | 33 | 4.6% |
| Scale: small landlord | 24,051 | 2.09 | $736 | 8 | 4.4% |
| Scale: large landlord | 30,710 | 2.00 | $761 | 2090 | 1.5% |
| Scale: medium landlord | 15,708 | 2.10 | $740 | 38 | 2.5% |

## 4.3 B2: Months-of-Arrears Regressions

Outcome: log(months) of back rent at filing. Models use `plaintiff_type` (Individual = reference) and `owner_type` (Person = reference) as categorical variables rather than separate binary dummies, which keeps the reference group explicit and tests all plaintiff/owner types jointly. All specs include year × tract fixed effects. Specs B2–B4 add `total_units_conglomerate_yr_net_parcel_bin` — a four-category measure of the landlord's portfolio units in *other* buildings (i.e. the conglomerate total excluding the evicted parcel itself): single-property, small ($ $10units), medium(10 - -49units), large(50+$ units). This isolates scale effects from building-level unit count already controlled by `num_units_bin`. Specs B3–B4 further add building type and unit-size controls. Spec B5

is the back-rent elasticity model with the same landlord scale control. Standard errors clustered at the building (PID) level.

**Key results:**

- **PHA plaintiff** files fastest: coefficient $\approx -0.41$ (B1) to $-0.48$ (B4), stable and large across all specifications. The effect is stronger than previously estimated, now that the sample includes single-unit rentals (PHA rarely evicts single-family tenants, sharpening the contrast).
- **Nonprofit plaintiff**: $\approx -0.14^{***}$ (B1), persisting at $\approx -0.10^{**}$ in B3. Nonprofits enforce noticeably faster than individual landlords even after controlling for building type and neighborhood.
- **For-profit corporate plaintiff**: $\approx -0.09^{***}$ (B1), shrinking to $\approx -0.02^{*}$ ($p = 0.036$) in B3 with full controls. The effect is now statistically significant — for-profit corporations file for modestly less arrears than individuals after conditioning on building characteristics and neighborhood.
- **Owner type** (Person = reference): For-profit corp and Nonprofit owners are indistinguishable from person owners ($p > 0.15$ in B3/B4). **PHA-owned buildings** have significantly *more* months of arrears at filing ($+0.09^{*}$, $p = 0.03$ in B3): tenants in PHA-owned buildings accumulate more debt before eviction is filed, consistent with PHA's more deliberate enforcement process.
- **High-filing buildings** evict sooner: main effect $-0.12$ (B3) to $-0.18$ (B4), highly significant throughout.
- **B4 interactions** (plaintiff type $\times$ high-filer building): All three corporate plaintiff types show significant positive interactions with the high-filer indicator (for-profit corp: $+0.085^{***}$; nonprofit: $+0.182^{***}$; PHA: $+0.326^{***}$). This means that in high-filer buildings, corporate and institutional plaintiffs file for *more* arrears relative to individual landlords in the same buildings — the speed advantage of institutional filers is attenuated in buildings where eviction is already a routinized practice.
- **Plaintiff-is-owner** near zero throughout ($p > 0.3$), consistent with management companies acting as agents rather than behaving differently from owner-filers.

## 4.4 B3: Back-Rent Elasticity

Outcome: log(total rent). Tests whether the back-rent amount scales proportionally with monthly rent ($\hat{\beta} = 1$) and how landlord type shifts the relationship.

The `log(ongoing rent)` elasticity is approximately **0.71** (significantly below 1 at any conventional level). This means higher-rent units are filed for proportionally *less* total back rent — consistent with landlords filing sooner at higher rent levels, possibly because vacancy and legal costs loom larger relative to each month of arrears. Among plaintiff types, only PHA shows a significant total-rent discount ($-0.32^{***}$), partly mechanical because surviving PHA cases are already above the \$300 rent floor and PHA policy triggers filing after one missed payment.

## 4.5 B4: Racial Disparity Regressions

These specifications use the imputed case-level defendant race file (`race_imputed_case_sample`) merged onto the winsorized case sample. The main race regressor is the **case-level Black probability**, not the building racial composition. All columns in Table 16 drop **PHA plaintiff cases**

Table 15: Back-Rent Months of Arrears Regressions

| Dependent Variable: | log_months | | | | |
|---|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) | (5) |
| *Variables* | | | | | |
| For-profit corporate plaintiff | -0.0811*** | -0.0447*** | -0.0186* | -0.0407*** | -0.0064 |
| | (0.0075) | (0.0109) | (0.0105) | (0.0125) | (0.0082) |
| Nonprofit plaintiff | -0.1427*** | -0.0590 | -0.0733* | -0.2221*** | -0.0707*** |
| | (0.0174) | (0.0372) | (0.0375) | (0.0387) | (0.0230) |
| PHA plaintiff | -0.4846*** | -0.5214*** | -0.5145*** | -0.5468*** | -0.4566*** |
| | (0.0185) | (0.0410) | (0.0398) | (0.0396) | (0.0250) |
| Log ongoing rent | -0.2499*** | -0.2334*** | -0.2598*** | -0.2629*** | -0.2593*** |
| | (0.0086) | (0.0086) | (0.0088) | (0.0088) | (0.0087) |
| For-profit corporate owner | | -0.0047 | 0.0355*** | 0.0338*** | |
| | | (0.0116) | (0.0118) | (0.0117) | |
| Nonprofit owner | | -0.0121 | 0.0177 | 0.0051 | |
| | | (0.0367) | (0.0360) | (0.0350) | |
| PHA owner | | 0.0364 | 0.0784* | 0.0797* | |
| | | (0.0471) | (0.0453) | (0.0448) | |
| Religious owner | | -0.2049** | -0.1446* | -0.1491* | |
| | | (0.0867) | (0.0810) | (0.0818) | |
| Trust owner | | -0.0886** | -0.0577 | -0.0570 | |
| | | (0.0381) | (0.0371) | (0.0376) | |
| Local government owner | | -0.0157 | 0.0710 | 0.0785 | |
| | | (0.0750) | (0.1092) | (0.1095) | |
| Financial intermediary owner | | -0.0838 | -0.0619 | -0.0623 | |
| | | (0.0512) | (0.0454) | (0.0461) | |
| High-filer building | | -0.1485*** | -0.1268*** | -0.1508*** | -0.1261*** |
| | | (0.0068) | (0.0070) | (0.0078) | (0.0070) |
| Other conglomerate units: small landlord | | -0.0048 | -0.0349*** | -0.0336*** | |
| | | (0.0089) | (0.0090) | (0.0090) | |
| Other conglomerate units: medium landlord | | 0.0262** | -0.0080 | -0.0061 | |
| | | (0.0120) | (0.0134) | (0.0133) | |
| Other conglomerate units: large landlord | | -0.0333** | -0.0414*** | -0.0398*** | |
| | | (0.0138) | (0.0144) | (0.0143) | |
| Plaintiff matches owner | | | -0.0162 | -0.0140 | |
| | | | (0.0109) | (0.0109) | |
| Building Black share | | | 0.0012 | 0.0019 | 0.0009 |
| | | | (0.0118) | (0.0117) | (0.0118) |
| 2-5 units | | | -0.0703*** | -0.0699*** | -0.0706*** |
| | | | (0.0102) | (0.0102) | (0.0102) |
| 21-50 units | | | -0.1562*** | -0.1520*** | -0.1605*** |
| | | | (0.0248) | (0.0247) | (0.0245) |
| 51+ units | | | -0.2494*** | -0.2435*** | -0.2446*** |
| | | | (0.0268) | (0.0267) | (0.0267) |
| 6-20 units | | | -0.1013*** | -0.0975*** | -0.0982*** |
| | | | (0.0236) | (0.0236) | (0.0235) |
| For-profit plaintiff x high-filer building | | | | 0.0407*** | |
| | | | | (0.0146) | |
| Nonprofit plaintiff x high-filer building | | | | 0.2108*** | |
| | | | | (0.0359) | |
| PHA plaintiff x high-filer building | | | | 0.3314*** | |
| | | | | (0.0432) | |
| Log conglomerate portfolio size | | | | | -0.0046* |
| | | | | | (0.0024) |
| *Fixed-effects* | | | | | |
| Year | Yes | Yes | Yes | Yes | Yes |
| Tract | Yes | Yes | Yes | Yes | Yes |
| Building type | | | Yes | Yes | Yes |
| *Fit statistics* | | | | | |
| Observations | 142,649 | 142,167 | 118,264 | 118,264 | 118,522 |
| $R^2$ | 0.11988 | 0.13116 | 0.14981 | 0.15076 | 0.14973 |
| Adjusted $R^2$ | 0.11746 | 0.12870 | 0.14683 | 0.14776 | 0.14682 |

*Clustered (PID) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 16: Back-Rent Elasticity Regression

| Dependent Variable: | Log total back rent |
|---|---|
| Model: | (1) |
| *Variables* | |
| Log ongoing rent | 0.7393*** |
| | (0.0087) |
| For-profit corporate plaintiff | -0.0202** |
| | (0.0102) |
| Nonprofit plaintiff | -0.0703* |
| | (0.0378) |
| PHA plaintiff | -0.5187*** |
| | (0.0394) |
| For-profit corporate owner | 0.0317*** |
| | (0.0116) |
| Nonprofit owner | 0.0165 |
| | (0.0365) |
| PHA owner | 0.1103** |
| | (0.0501) |
| Religious owner | -0.1463* |
| | (0.0822) |
| Trust owner | -0.0575 |
| | (0.0369) |
| Local government owner | 0.0899 |
| | (0.1098) |
| Financial intermediary owner | -0.0599 |
| | (0.0458) |
| High-filer building | -0.1262*** |
| | (0.0070) |
| Log conglomerate portfolio size | -0.0086*** |
| | (0.0029) |
| Building Black share | 0.0015 |
| | (0.0117) |
| *Fixed-effects* | |
| Year | Yes |
| Unit-count bin | Yes |
| Tract | Yes |
| Building type | Yes |
| *Fit statistics* | |
| Observations | 118,522 |
| $R^2$ | 0.36508 |
| Adjusted $R^2$ | 0.36287 |

*Clustered (PID) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

**and PHA-owned parcels**. Column (1) includes only `case_p_black` with **no** year, tract, or building-type fixed effects. Column (2) adds the full fixed effects and controls. Column (3) adds the plaintiff-type interaction terms on top of column (2). Column (4) goes back to column (2) but removes tract fixed effects. The merged race file covers 153,392 cases in the winsorized sample; `Building Black share` remains in the table as a contextual control.

Table 17: Racial Disparity Regressions

| Dependent Variable: | log_months | | | |
|---|---|---|---|---|
| Model: | (1) | (2) | (3) | (4) |
| *Variables* | | | | |
| Constant | 0.7371*** | | | |
| | (0.0149) | | | |
| Case Black probability | 0.1039*** | -0.0036 | -0.0181* | 0.0040 |
| | (0.0189) | (0.0077) | (0.0107) | (0.0087) |
| For-profit corporate plaintiff | | -0.0285*** | -0.0439*** | -0.0441*** |
| | | (0.0104) | (0.0132) | (0.0154) |
| Nonprofit plaintiff | | -0.0952*** | -0.0992* | -0.0980** |
| | | (0.0355) | (0.0561) | (0.0386) |
| For-profit corporate owner | | -0.0006 | -0.0009 | 0.0031 |
| | | (0.0112) | (0.0112) | (0.0188) |
| Nonprofit owner | | -0.0275 | -0.0278 | 0.0185 |
| | | (0.0342) | (0.0345) | (0.0393) |
| Religious owner | | -0.1976** | -0.2019** | -0.1826* |
| | | (0.0833) | (0.0832) | (0.1108) |
| Trust owner | | -0.0915*** | -0.0915** | -0.0931* |
| | | (0.0353) | (0.0355) | (0.0515) |
| Local government owner | | 0.0394 | 0.0399 | 0.0231 |
| | | (0.1318) | (0.1320) | (0.1604) |
| Financial intermediary owner | | -0.0958** | -0.0954** | -0.0575 |
| | | (0.0477) | (0.0478) | (0.0415) |
| Plaintiff matches owner | | -0.0021 | -0.0026 | 0.0022 |
| | | (0.0112) | (0.0112) | (0.0160) |
| Log ongoing rent | | -0.3117*** | -0.3116*** | -0.3208*** |
| | | (0.0086) | (0.0086) | (0.0106) |
| Building Black share | | -0.0167 | -0.0142 | 0.0125 |
| | | (0.0120) | (0.0122) | (0.0120) |
| Case Black probability x for-profit plaintiff | | | 0.0264* | |
| | | | (0.0136) | |
| Case Black probability x nonprofit plaintiff | | | 0.0089 | |
| | | | (0.0523) | |
| *Fixed-effects* | | | | |
| Year | | Yes | Yes | Yes |
| Tract | | Yes | Yes | |
| Building type | | Yes | Yes | Yes |
| *Fit statistics* | | | | |
| Observations | 113,119 | 113,118 | 113,118 | 113,119 |
| $R^2$ | 0.00506 | 0.14517 | 0.14524 | 0.10863 |
| Adjusted $R^2$ | 0.00505 | 0.14210 | 0.14216 | 0.10841 |

*Clustered (PID) standard-errors in parentheses*
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 18: Key findings summary

| Finding | Evidence |
|---|---|
| Neighborhood sorting is the dominant source of filing rate variation | $\Delta R^2$(tract FE) $\approx$ 0.082 vs. $\Delta R^2$(landlord) $\approx$ 0.009 |
| Landlord type (owner type + entity portfolio) explains $\approx$0.9% of cross-sectional variation | M4 incremental $R^2$ from M3; robust to PPML |
| Conglomerate portfolio (M5) explains similar share to entity portfolio (M4) | M5 $\Delta R^2$ from M3 comparable to M4; conglomerate portfolio in B5 reg. |
| PHA files fastest: 41–48% fewer log-months, highly stable across all specs | B1: $-0.41^{***}$; B4: $-0.48^{***}$; 6,317 cases survive \$300 rent floor |
| Nonprofits file faster; for-profit corp effect significant with full controls | Nonprofit B3: $-0.10^{**}$; for-profit corp B3: $-0.02^{*}$, $p = 0.036$ |
| High-filing buildings evict sooner; corporate types partially offset this in high-filer bldgs | B4 interaction for-profit$\times$high-filer: $+0.085^{***}$; nonprofit: $+0.182^{***}$ |
| PHA-owned buildings accumulate more arrears before filing ($+0.09^{*}$) | B3 owner_typePHA: $+0.090^{*}$, $p = 0.030$ |
| Back-rent elasticity w.r.t. monthly rent = 0.71 (significantly below 1.0) | Elasticity $\approx$ 0.71; 95% CI excludes 1.0 |
| Plaintiff = building owner in only 15.8% of cases | Name lookup exact match: 29.1%; entity match: 15.8% |

# 5 Summary

## 5.1 Caveats

1. **Plaintiff lookup rate (29.1%):** Most eviction plaintiffs are management companies not present in RTT or OPA by exact name. Fuzzy-name matching (Phase 3, deferred) could substantially improve coverage.

2. **OPA fallback ownership (63.4% of PID-years):** Buildings with no RTT history inherit their 2024 OPA owner for all years, introducing attenuation bias if ownership changed during 2011–2019.

3. **Improved entity reconciliation (Phase 2/3):** As of the 2026-03-10 pipeline update, persons and trusts are grouped by (name, PID) rather than name alone. Phase 2 now uses name+address matching (not pure address sharing) to prevent registered-agent and government-hub explosion. Phase 3 adds a name-similarity gate preventing tax-exempt government conveyances from polluting private landlord conglomerates. These improvements make `portfolio_bin_conglomerate_yr` (M5/B5) a more reliable measure of true landlord scale than entity-level portfolio (M4).

4. **Racial disparity specs (B4):** Now active. The case-level defendant race file (`race_imputed_case_sample`) merged onto 153,392 cases in the winsorized sample. The current Table 16 specs drop PHA plaintiff cases and PHA-owned parcels, leaving 113,118–113,119 observations in the race-regression sample depending on the specification.